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Coarse-to-fine-grained insulator defect detection using two-stage deep learning framework

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Introduction: In power transmission and power distribution systems, insulators are the fundamental components which provide mechanical support and electrical isolation between conductors and supporting towers. However, these insulators are continuously exposed to extreme weather conditions, mechanical stress, pollution and dust, which can lead to surface damage and abnormal operation.

Method: Conventionally, manual on-site inspection is performed to identify faulty insulators. However, this method is labor intensive, risky, and reactive. With the advancement in unmanned aerial vehicle (UAV) technology and deep learning method, automatic visual inspection has replaced the conventional methods. However, automatic visual inspection from UAV images is challenging task due to complex background and small size of the insulators. To address these challenges, we proposed a framework that consists of two stages. In the first stage, you only look once (YOLO)-based is used for extracting candidate region of insulator to get coarse class prediction. During the second stage, the candidate regions are then cropped from the image into image patches and fed to a lightweight patch-level classifier. In order to enhance inter-class separability, hard-pair contrastive classification loss is proposed by adding hard-pair contrastive term to the cross-entropy loss in the embedding space.

Discussion: The proposed technique is evaluated using an open-source insulator defect image dataset. The experiment results demonstrate the superior performance of the proposed framework compared with the YOLO variants and representative two-stage detectors.

KEYWORDS
deep learning, hard-pair contrastive classification loss, insulator defect detection, UAV, YOLO - you only look once

1 Introduction

Insulators are the basic components of power transmission and distribution networks (Nzenwa and Adebayo, 2019). The insulators provide mechanical support and electrical isolation between conductors and towers. To maintain healthy power system and to prevent flashovers, leakage current, and other service interruptions, it is important to ensure the reliable operation of these insulators (Salem et al., 2022). However, these insulators are continuously exposed to harsh weather, dust, pollution, aging, etc., in the outdoor environment. These factors adversely affect the quality of insulators and lead to several types of defects, including, glaze flash over damage, broken shells, surface dust, and other abnormal conditions. The performance of power transmission degrades gradually and may increase the risk of power outages if these defects are not rectified at the early

stage (Hatziargyriou et al., 2020). Furthermore, without lack of early-stage defect detection may lead to high maintenance costs later (Divya et al., 2023). Therefore, insulator defect detection at the early stage is an important task in power line inspection and smart grid maintenance (Liu et al., 2023).

Traditionally, insulator inspection is usually performed manually by a field expert (Jain et al., 2024). In many cases, the inspection is usually carried out after visible performance degradation or power outages (Han et al., 2019). This makes the whole inspection process reactive rather than preventive. This conventional method has several limitations. First, manual inspection is a time consuming, risky, and labor-intensive job. Second, maintenance personnel may need to work near high-voltage equipment which exposes them to serious safety risks. Third, minor insulator damage, such as glaze flashover marks, dust accumulation, small broken insulators, can be difficult to distinguish visually. This may lead to several missed detection which cause delays in maintenance decisions.

With the advancement of unmanned aerial vehicle (UAV) technology and machine vision algorithms, traditional methods have been replaced by automatic visual inspection methods Jordan et al. (2018). Automatic visual inspection employ deep learning based image analysis methods to automatically analyse high resolution images captured by the UAVs Liu et al. (2022). Several studies have been reported in the literature that show the effectiveness of automatic visual inspection in different domains. For example, Ye et al. (2024) proposed enhanced YOLOv5s model for UAV-based wind turbine blade crack detection. Kalfarisi et al. (2020) proposed two deep learning approaches for crack detection and segmentation. Several studies have been proposed (Shanti et al., 2022; Alateeq et al., 2023; Liang et al., 2023) for construction site monitoring, and (Vega Díaz et al., 2020) for solar panel inspection.

Insulator inspection is a challenging task due to the following reasons: (1) UAV images generally have complex backgrounds, large-scale variations, occlusions, and illumination changes; (2) the size of some defects may be small and occupy small portion in the whole image, which makes it difficult for the detectors to accurately detect them; and (3) different defect categories may visually and structurally look similar which can confuse the detectors. For example, two different categories from insulator defect dataset (IDID) are illustrated in Figure 1. As shown in figure, both categories appear very similar, with a well-preserved disk shape, and no visible shell damage. The main difference between these categories lies in fine-grained local appearance features which are often mixed with background information causing the detectors to confuse the two categories.

To address these challenges, several methods are reported in literature. Zhao et al. (2021) proposed an improved Faster R-CNN model for insulator recognition and fault detection. Qiu et al. (2022) developed an improved lightweight YOLOv4 model for insulator defect detection in transmission lines. He et al. (2024) proposed YOLOv8s-SwinT, where Swin Transformer, improved BiFPN, and a detection layer (for small targets) were integrated to enhance defect detection. Chai et al. (2025) proposed FPFS-YOLO based on YOLO11n for insulator defect detection. Although, these models improve the detection accuracy, however, most of them still rely on single-stage detection which limits their ability to handle fine-grained class confusion.

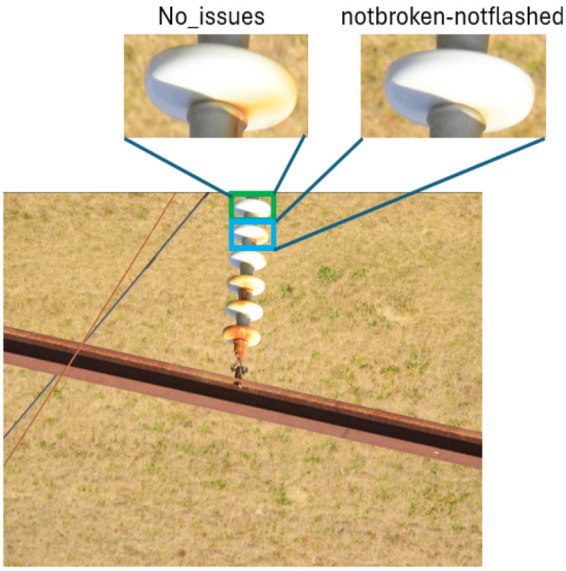


FIGURE 1
Visual comparison of two visually similar but different categories from IDID dataset.

To overcome the aforementioned challenges, a coarse-to-fine-grained two-stage deep learning approach was proposed for insulator defect detection. Generally, the framework consists of two stages. In the first stage, a YOLOv8m-based detector is used to locate the candidate insulator regions in the UAV image. In the second stage, the detected regions are then cropped into image patches and fed to a lightweight patch-level classifier for fine-grained defect classification. To further improve the separability of visually similar classes, a hard-pair contrastive classification loss is introduced. Compared with existing methods, the proposed framework offers the following contributions:

In addition to conventional cross-entropy supervision, HPCCL pulls embeddings from the same hard class closer while pushing embeddings from the confusing classes farther apart, resulting in a more discriminative feature space.

1. A two-stage detection and classification framework is introduced to decouple the defect localization from fine-grained detection task. This approach helps to reduce background effects and highlights the local properties of visually similar defects.
2. A Hard-Pair Contrastive Classification Loss (HPCCL) is introduced by adding hard-pair contrastive term to the cross-entropy loss in the embedding space. This loss improves the discriminative features space between visually similar categories.
3. Extensive experiments are performed to compare the proposed method with single-stage YOLO variants and representative two-stage detectors. Experiment results demonstrate the effectiveness of proposed framework on challenging benchmark dataset.

The rest of the paper is organized as follows: the next section discusses the related work, the section following the next discusses the proposed framework, experiment results are discussed in the penultimate section, and the final section concludes the paper.

2 Related work

Automatic visual inspection of insulator has received attention from the research community. Several methods and approaches have been developed to early detect the insulator defects to ensure healthy power transmission.

2.1 CNN-based two-stage methods

Liu et al. (2023) presented a comprehensive survey on insulator defect detection where the authors discussed that the CNN-based object detection models and transformer-based deep learning models have shown better performance than the conventional handcrafted image processing. Tao et al. (2018) proposed a deep learning model to detect defects on power-line insulators in aerial images. Jiang et al. (2019) used a multi-level perception and ensemble-learning strategy to detect insulator defects. Adou et al. (2019) also studied the application of deep learning in the field of insulators fault detection, and Liu et al. (2021) proposed a deep learning model for the aerial image-based detection of insulators faults on high-voltage transmission poles. Although these methods demonstrated the superiority of CNN-based inspection methods, however, many of them focus on limited defect categories. The performance of these models may degrade in complex scenes, where the size of defects is small, partially occluded, and have similar visual appearances.

Due to strong localization capability, several two-stage frameworks are also proposed from insulator defect detection. Liao and Yang (2019) proposed an improved Faster R-CNN model for insulator defect detection in aerial images. The authors used ResNet-101 as backbone for feature extraction and employ soft non-maximum suppression to suppress the overlapping insulator bounding boxes. Wen et al. (2021) proposed faster R-CNN which consists of CME-CNN and Exact R-CNN. The framework integrates feature pyramid network (FPN) and utilizes mask extraction to suppress complex background interference for insulator defect detection. These methods utilize region proposal mechanism to generate candidate regions and then employ different refinement methods for precise localization. However, these methods are computationally expensive and not suitable for real-time UAV deployment.

2.2 YOLO-based methods

Many studies adopted YOLO-based detector for insulator defect detection in real time. These YOLO-based models are single-stage detectors and are popular due to their optimum balance between the detection accuracy and inference time. An enhanced YOLOv4 lightweight model is proposed by Qiu et al. (2022) for insulator defect detection. The framework utilizes GraphCut-based data augmentation techniques and employs transfer learning to achieve high accuracy in real time. Similarly, Ding et al. (2022) and Wang et al. (2023) proposed an enhanced YOLOv5 model for real-time insulator defect detection. Zheng et al. (2022) proposed an improved YOLOv7-based model for detecting insulator defects in complex background. The framework integrates K-means++ anchor optimization, HorBlock modules, SIoU, and enhanced NMS to improve detection accuracy of small defects. Liu et al. (2022) proposed cascaded YOLO models that combines enhanced YOLOv3-based detector with an improved YOLOv4-tiny for insulator defect detection in aerial images.

These methods demonstrate the effectiveness of YOLO-based detectors for fast insulator defect detection in UAV images, however, these methods still perform coarse localization and classification jointly which limits their capability to distinguish fine-grained and visually confusing defect classes.

2.3 Attention-based methods

The performance of YOLO-based models improved further in the recent studies by integrating attention mechanism, multi-scale feature fusion strategies, and transformer modules. Wang et al. (2023) proposed ML-YOLOv5 for insulator defect detection that integrates depthwise separable convolutions, an improved C2f_DG module, and an enhanced feature pyramid network to achieve high accuracy in real-time. Wei et al. (2024) improved YOLOv5s by employing CBAM (Woo et al., 2018) attention mechanism which improves feature extraction and achieve high accuracy. He et al. (2024) proposed YOLOv8s-SwinT framework which integrates Swin (shifted window) transformer and an improved BiFPN to improvise tiny-defect detection. Xu and Tang (2025) proposed MAP-YOLOv8 that introduces novel MAP-CA attention mechanism and super-resolution enhancement to improve detection under low-resolution imaging conditions. Chai et al. (2025) proposed FPFS-YOLO model which integrates FasterNet, ParNet attention mechanism, and lightweight detection head into YOLO11n.

2.4 Transformer-based methods

Transformer-based models were also developed for insulator defect detection. Transformer-based models can efficiently model global contextual information and long-range dependencies which provides support in object detection in cluttered and complex UAV images. Li et al. (2024) proposed DEtection TRansformer (DETR)-based framework which incorporates multi-scale features from the backbone and employs self-attention mechanism for accurate insulator defect detection. Similarly, Cheng and Liu (2022) introduces DETR-based framework for insulator defect detection in UAV images. He et al. (2024) proposed a framework that integrates Swin transformer with an improved BiFPN for tiny insulator defect detection. Xi et al. (2023) proposed a framework that enhances Faster R-CNN by integrating Swin transformer as backbone for insulator defect detection. The framework employs

deformable receptive field block which improves the contextual feature extraction.

3 Methodology

This section provides the details of proposed two-stage framework for insulator defect detection. The framework consists of two complementary stages: In the first stage, YOLOv8m-based detector takes the input image and predicts the coarse locations, categories, and scores of the insulator objects. The primary goal of the first stage is the object localization. In the second stage, patches corresponding to the detected regions are cropped and provided as input to a lightweight patch-level classifier. The patch-level classifier is used to perform fine-grained classification of the patches. To address inter-class similarity problem between categories, (`No_issues`, and `notbroken-notflashed`), a hard-pair contrastive loss is introduced to enforce the class separability. Detailed architecture of proposed framework is illustrated in Figure 2.

3.1 YOLOv8m-based coarse detection

In the first stage, a YOLOv8m detector is trained using the original full images for the coarse prediction of the location and categories of the insulator objects. In this stage, YOLOv8m is selected due to its multi-scale architecture which enables the model to detect different size regions under various distances, view point, and image resolution. Furthermore, inspection scenarios require reliable performance with computational efficiency in real time which make it a suitable choice for the current study.

YOLOv8m model is trained on labeled dataset by matching its predicted boxes with the ground-truth boxes. Let the labeled training dataset be represented as $\mathcal{D} = \{(I_n, \mathcal{G}_n)\}_{n=1}^{N}$,, where $I_n \in \mathbb{R}^{H\times W\times 3}$ denotes the n-th input image and $\mathcal{G}_n$ denotes the corresponding set of ground-truth annotations. Images in the dataset may contain multiple annotated insulator defect regions. The ground-truth annotation set for image I_n is defined as $\mathcal{G}_n = \{(g_{nj}, y_{nj})\}_{j=1}^{M_n}$,, where M_n is the number of annotated objects in image I_n, g_{nj} is the ground-truth bounding box of the j-th object, and $y_{nj} \in \{0, 1, 2, 3\}$ is its class label. The detector is trained using the standard YOLO loss function as in Equation 1.

$$\mathcal{L}_{YOLO} = \lambda_{box}\mathcal{L}_{box} + \lambda_{dfl}\mathcal{L}_{dfl} + \lambda_{cls}\mathcal{L}_{cls}, \tag{1}$$

where $\mathcal{L}_{box}$ penalizes inaccurate bounding-box predictions, $\mathcal{L}_{dfl}$ represents the Distribution Focal Loss addressing boundary ambiguity around object edges, and $\mathcal{L}_{cls}$ trains the detector to assign preliminary class labels. The coefficients λ_{box}, λ_{dfl}, and λ_{cls} control the relative contribution of each loss term.

After training, YOLOv8m detector is used to predict coarse bounding boxes. These bounding boxes are utilized to extract the patches from the input image and later on these patches are utilized to trained fine-grained patch-level classifier for final output.

Let I_n represent an image provided as input to the YOLOv8m model, and $\hat{\mathcal{B}}_n$ is a set of candidate detections as: $\hat{\mathcal{B}}_n = F_\theta(I_n) = \{(\hat{b}_{ni}, \hat{s}_{ni}, \hat{c}_{ni})\}_{i=1}^{K_n}$,, where K_n is the number of predicted candidate boxes, $\hat{b}_{ni}$ is the predicted bounding box, $\hat{s}_{ni}$ is the detection confidence score, and $\hat{c}_{ni}$ is the preliminary class label predicted by the YOLO detector.

At inference time, YOLOv8m may generate multiple overlapping predictions for the same object. Therefore, non-maximum suppression (NMS) is applied to remove redundant detections. The refined set of detections is defined as in Equation 2

$$\mathcal{B}_n^* = \mathrm{NMS}(\hat{\mathcal{B}}_n, \tau), \tag{2}$$

where τ is the intersection over union (IoU) threshold used for suppression. The remaining detections $\mathcal{B}_n^*$ are then forwarded to the patch extraction module.

The YOLO detector is supervised by class-aware labels, but the predicted category is considered as coarse labels, while the bounding box and detection confidence are kept for the next stage, and the final fine-grained class label is determined by the patch-level classification.

3.2 Patch extraction for fine-grained classification

After receiving the set of detections, each bounding box is used to crop a corresponding image patch from the input image.

Let $b_{ni}^* \in \mathcal{B}_n^* = (x_{ni}^1, y_{ni}^1, x_{ni}^2, y_{ni}^2)$. is the bounding box. Each bounding box is padded to preserve local context around the detected region. The width and height of the box are computed as in Equation 3

$$w_{ni} = x_{ni}^2 - x_{ni}^1, \quad h_{ni} = y_{ni}^2 - y_{ni}^1. \tag{3}$$

The padded coordinates are defined as in Equations 4, 5

$$\tilde{x}_{ni}^1 = x_{ni}^1 - \alpha w_{ni}, \quad \tilde{y}_{ni}^1 = y_{ni}^1 - \alpha h_{ni}, \tag{4}$$

$$\tilde{x}_{ni}^2 = x_{ni}^2 + \alpha w_{ni}, \quad \tilde{y}_{ni}^2 = y_{ni}^2 + \alpha h_{ni}, \tag{5}$$

where α is the padding ratio. In this study, $\alpha = 0.10$ is used to include 10% contextual information around the detected region. The cropped patch P_{ni} is then resized equal to size of the input of the patch-level classifier. The patch-level training dataset, $\mathcal{D}_{patch}$ is created after extracting patches from images belonging to training subset as defined in Equation 6.

$$\mathcal{D}_{patch} = \{(P_m, y_m)\}_{m=1}^{M}, \tag{6}$$

where P_m denotes the cropped image patch and y_m is its corresponding ground-truth class label.

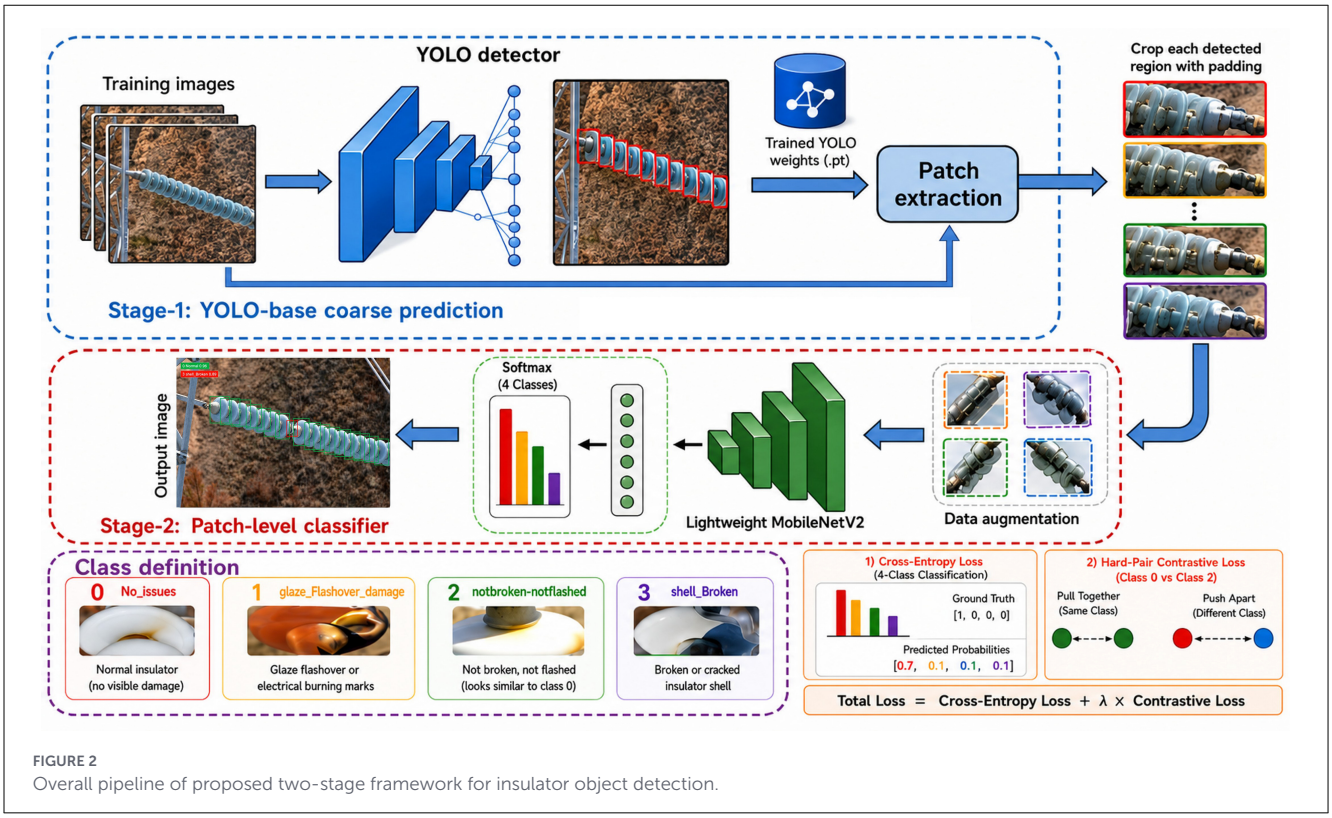


FIGURE 2
Overall pipeline of proposed two-stage framework for insulator object detection.

3.3 Stage 2: patch-level MobileNetV3 classifier

The second stage is the patch-level classifier which classifies the patches extracted in the previous stage. Since the classifier utilizes image patches instead of processing the whole image, therefore, it is able to extract local features of the object.

In this stage, lightweight MobileNetV3-based classifier is used to refine the preliminary category predictions. This model is chosen as it provides good balance between the classification accuracy and computational complexity. MobileNetV3 consists of three main components, feature extractor, embedding projection layer, and classification head. Let P_m be an input patch to MobileNetV3 classifier. The classifier extract hierarchy features from the input image and maps the input patch into a feature representation as $f_m = \Phi(P_m)$, where $\Phi(\cdot)$ denotes the MobileNetV3 feature extractor. The extracted feature is then passed through an embedding projection module: which converts the feature maps into embedding vector as $z_m = \Psi(f_m)$, where $z_m \in \mathbb{R}^d$ is the embedding vector and d is the embedding dimension. The embedding vector is then normalized as $\bar{z}_m = \frac{z_m}{\|z_m\|_2}$. and provided as input to the classification head which produces the class logits as $q_m = W\bar{z}_m + b$, where W and b are learnable classifier parameters. The predicted class probability distribution is obtained using the softmax function in Equation 7:

$$p_m(k) = \frac{\exp(q_m^k)}{\sum_{r=0}^{C-1} \exp(q_m^r)}, \tag{7}$$

where $C = 4$ is the number of classes and $p_m(k)$ is the predicted probability of class k.

The patch-level classifier predicts the final class label as in Equation 8.

$$\hat{y}_m = \arg\max_k p_m(k). \tag{8}$$

As this stage operates on the image patches, therefore, it improves fine-grained classification. This stage reduces the influence of background information and allows the classifier to focus on small visual differences between the object categories.

3.4 Hard-pair contrastive classification loss

The patch-level classifier is optimized using a combined loss function, named the *Hard-Pair Contrastive Classification Loss* (HPCCL). This loss function combines the conventional cross entropy loss with hard-pair contrastive loss. The motivation behind HPCCL is that cross-entropy loss function predicts the correct

category for each cropped patch. However, this loss treats all the classes equally. Due to this reason, cross entropy loss function leads to confusion between `No_issues` and `notbroken-notflashed`. Therefore, an additional hard-pair contrastive loss is introduced. The hard-pair contrastive loss enhances the separability of visually similar classes in the embedding space.

The proposed Hard-Pair Contrastive Classification Loss is formulated as in Equation 9.

$$\mathcal{L}_{HPCCL} = \mathcal{L}_{CE} + \lambda \mathcal{L}_{HPC}, \tag{9}$$

where $\mathcal{L}_{CE}$ is the cross-entropy loss and $\mathcal{L}_{HPC}$ hard-pair contrastive loss, λ is a weighting coefficient that controls the contribution of the hard-pair contrastive term. A positive value of λ adds feature-level supervision and encourages stronger separation between the hard classes. This study uses λ= 0.2. This value is empirically determined through experiments.

For a patch P_m with ground-truth label y_m, the patch classifier predicts a class probability distribution $p_m(k)$, where $k \in \{0, 1, \ldots, C-1\}$ and C is the total number of classes. The standard cross-entropy loss is defined in Equation 10.

$$\mathcal{L}_{CE} = -\frac{1}{M} \sum_{m=1}^{M} \sum_{k=0}^{C-1} \Omega_{m,k} \log \left(p_m(k) + \epsilon \right), \tag{10}$$

where C is the number of classes, M is the number of cropped patches, $p_m(k)$ is the predicted probability that patch P_m belongs to class k, and $\Omega_{m,k}$ is the one-hot encoded ground-truth label and defined by Equation 11. Specifically,

$$\Omega_{m,k} = \begin{cases} 1, & \text{if } y_m = k, \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

The small constant ϵ is added for numerical stability to avoid taking the logarithm of zero.

To compute hard-pair contrastive term, let $\bar{z}_i$ and $\bar{z}_j$ be two L_2-normalized embeddings generated by the embedding layer of the classifier. The cosine distance between the two embeddings is computed as in Equation 12

$$d(\bar{z}_i, \bar{z}_j) = 1 - \bar{z}_i^T \bar{z}_j. \tag{12}$$

Only samples whose labels belong to the hard class set $\mathcal{C}_h$ are used in the contrastive term. The hard-sample index set is defined as in Equation 13

$$\mathcal{H} = \{m \mid y_m \in \mathcal{C}_h\}. \tag{13}$$

From $\mathcal{H}$, positive pairs are formed using samples that belong to the same class as defined by Equation 14:

$$\mathcal{P} = \{(i,j) \mid y_i = y_j,\ i,j \in \mathcal{H},\ i \neq j\}. \tag{14}$$

Negative pairs are formed using samples that belong to different classes within the hard class set as in Equation 15:

$$\mathcal{N} = \{(i,j) \mid y_i \neq y_j,\ i,j \in \mathcal{H}\}. \tag{15}$$

The hard-pair contrastive loss is formulated as in Equation 16

$$\mathcal{L}_{HPC} = \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} d(\bar{z}_i, \bar{z}_j) + \frac{1}{|\mathcal{N}|} \sum_{(i,j)\in\mathcal{N}} \max(0, \mu - d(\bar{z}_i, \bar{z}_j)), \tag{16}$$

where μ is the contrastive margin. The first term pulls embeddings of samples from the same hard class closer together, while the second term pushes embeddings of samples from different hard classes apart when their distance is smaller than the margin. As a result, the classifier learns a more discriminative feature space for categories that are visually similar and difficult to separate. This study maintains the value of μ = 0.5 which is determined through experiments by keeping all the other training parameters fixed.

3.5 Two-stage inference and final prediction

During inference, the input image is first processed by the trained YOLOv8m detector. After NMS, each retained bounding box is cropped from the image and passed to the patch classifier. The YOLOv8m detector provides the localization output, while the patch classifier provides the final class label.

For each retained detection b_i^*, the classifier predicts a probability distribution over the four classes as defined in Equation 17:

$$p_i^{cls} = \{p_i^{cls}(0), p_i^{cls}(1), p_i^{cls}(2), p_i^{cls}(3)\}. \tag{17}$$

The final class label is determined as in Equation 18

$$c_i^{final} = \arg\max_k p_i^{cls}(k). \tag{18}$$

The final confidence score combines the YOLOv8m detection confidence and the classifier confidence as defined in Equation 19:

$$s_i^{final} = s_i^{det} \times \max_k p_i^{cls}(k), \tag{19}$$

where s_i^{det} is the YOLO confidence score and $\max_k p_i^{cls}(k)$ is the confidence of the patch classifier. The final output of the proposed framework is defined in Equation 20

$$\mathcal{O} = \{(b_i^*, c_i^{final}, s_i^{final})\}_{i=1}^{R}, \tag{20}$$

where R is the number of retained detections after NMS.

This two-stage inference strategy allows the framework to benefit from both localization and fine-grained recognition.

In case, the coarse YOLO prediction and the patch-level prediction disagree, the patch-level prediction takes precedence because the second stage is specifically optimized to resolve fine-grained inter-class confusion.

4 Experiment results

In this section, the performance of the proposed framework is evaluated and its performance is compared with single- and two-stage detector. The proposed framework is implemented in Python version 3.11 on Ubuntu 20.04 with Ultralytics 8.4.26 library, PyTorch: 2.11.0, and CUDA: 12.8. The experiments are performed on NVIDIA GeForce RTX 2070 GPU with 8 GB memory. The framework was trained for 100 epochs with initial learning rate 0.01 and parameters of the model are optimized by using Adam optimizer with weight decay. A batch size of 8 was used during training, and the input images were resized to 960 × 960 pixels.

4.1 Data description

Insulator Defect Image Dataset (IDID) was used to evaluate the effectiveness of the proposed framework and compare its performance with existing approaches. The dataset is collected by Kulkarni (2022) which consists of 1,600 drone view images with four categories : `No_issues`, `glaze_Flashover_damage`, `notbroken-notflashed`, `shell_Broken`. A summary of the dataset is provided in Table 1. As illustrated in the Table 1, the number of images varies significantly across the categories and so the number of instances. Furthermore, the average defect area also differs significantly among the categories. These characteristics make the dataset more challenging for training the model and increase the risk of biased learning toward dominant classes. Few sample frames from the dataset are illustrated in Figure 3.

To address the class imbalance problem, targeted augmentation technique was applied to increase the number of weak categories. Augmentation technique was not applied for test subsets. All augmented versions of an image remained in the same training partition as their corresponding original image. Therefore, augmented copies of a source image could not appear in test sets. This process upsamples minority classes during training and avoids information leakage and guarantees that the evaluation metrics reported during the evaluation are derived from unseen samples.

It is to be noted that the original images were split into mutually exclusive training, and test sets with a ratio of [70:30] before any data augmentation was applied. The split was done at the level of the original image, so that every image source and all annotations related to it were assigned to one subset only. Therefore, no original images or copies of images were shared across the training set, and test set. All experiments were conducted with the same test set, which was used for model validation and final testing, respectively.

4.2 Evaluation metrics

To evaluate the performance of proposed framework, standard evaluation metrics, precision, recall, F1-score, and mean average precision at an IoU threshold of 0.5 (mAP@50) are used. These evaluation metrics are commonly used to evaluate the performance of object detectors because these metrics measure both the correctness of predicted classes as well as the quality of object localization.

A predicted bounding box is considered as true positive if it matches the ground-truth class and its IoU with the corresponding ground-truth bounding box being greater than or equal to 0.5, otherwise it will be treated as false negative.

The IoU between a predicted bounding box B_p and a ground-truth bounding box B_g is computed as $IoU = \frac{|B_p \cap B_g|}{|B_p \cup B_g|}$, where $|B_p \cap B_g|$ represents the area of overlap between the predicted and ground-truth boxes, and $|B_p \cup B_g|$ represents their union area.

Precision measures the proportion of correctly detected positive samples among all predicted positive samples and formulated as $Precision = \frac{TP}{TP+FP}$, where TP denotes true positives and FP denotes false positives.

Recall measures the proportion of correctly detected positive samples among all actual positive samples and formulated as $Recall = \frac{TP}{TP+FN}$, where FN denotes false negatives. A high recall value indicates that the model misses fewer defect instances.

The F1-score provides a balanced measure between precision and recall and formulated as $F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$.

Average precision (AP) summarizes the precision–recall curve for a given class. It is obtained by integrating precision over recall as in Equation 21.

$$AP = \int_0^1 P(R)\, dR, \tag{21}$$

where $P(R)$ denotes precision as a function of recall. The mean average precision at an IoU threshold of 0.5 is computed by averaging the AP values over all C object classes and formulated as in Equation 22.

$$mAP@50 = \frac{1}{C} \sum_{c=1}^{C} AP_c^{IoU=0.5}, \tag{22}$$

where $AP_c^{IoU=0.5}$ is the average precision of class c when detections are considered correct at $IoU \geq 0.5$. In this study, $C = 4$, corresponding to the four insulator categories.

4.3 Comparison with single-stage YOLO variants and defect confusion suppression configurations

To evaluate the effectiveness of the proposed two-stage framework, several single-stage YOLO-based detectors and their modified variants were compared. This experiment divides the single-stage detectors into the following two groups.

The first group consists of standard single-stage detectors. These single-stage detectors include, YOLO11n, YOLO11s, YOLO8s, YOLO8m, and YOLO8n. These single-stage detectors predict the bounding box and class label in a single forward pass and provide good trade-off between accuracy and computational complexity for the detection tasks.

The goal of the second group is to evaluate Defect Confusion Suppression (DCS) which consist of YOLO11n as base network with three different configurations. The rationale for using these configurations is to understand whether the targeted architectural

TABLE 1 Category-wise distribution and annotation-size ranges of the original insulator dataset.

Class ID	Category	Images	Instances	Area (px^2)	Width (px)	Height (px)
0	No_issues	1,532	13,060	16,376–2,124,803	92–1,805	83–1,997
1	glaze_Flashover_damage	824	2,564	19,894–3,200,042	171–1,792	98–2,085
2	notbroken-notflashed	40	304	55,575–314,161	271–680	171–462
3	shell_Broken	812	1,180	4–1,335,913	2–1,553	2–1,297

The category-wise image counts overlap because a single image may contain instances from multiple categories.



FIGURE 3
Illustrates random samples from the dataset. The categories are color coded in red, green, yellow, and cyan. Red color represents the shell_Broken, green color represents No_issues, yellow represents glaze_Flashover_damage, and cyan represents notbroken-notflashed.

changes, particularly inserting attention modules in lightweight YOLO11n, can improve the performance for insulator defect detection. For this purpose, the following three configurations are generated.

1. DCS-Attn: in this configuration, YOLO11n is modified by inserting lightweight attention modules after the P3 and P4 backbone features. This modification is aimed to improve feature discrimination.
2. DCS-BHM: in this configuration, the model from the DCS-Attn is fine-tuned using a balanced hard-mining dataset. This configuration mines hard negatives and missed hard positives from the training split.
3. DCS-HN: in this configuration, the DCS-Attn model is fine-tuned using duplicated hard-negative samples, especially cases where class 0 (No_issues) is confused with class 2 (notbroken-notflashed).

The results are reported in Table 2. From the Table 2, it is indicated that among the standard YOLO variants, YOLO8m achieves good performance with a precision of 0.88, recall of 0.91, F1-score of 0.90, and mAP@50 of 0.9. YOLOv8s and YOLO11s also achieve comparable results with F1-scores of 0.89 and 0.88, respectively. On the other hand, the lighter models YOLO11n and YOLOv8n achieve relatively lower performance compared to their larger counterparts. This is due to reason that these models have limited feature extraction capacity due to their lightweight architecture.

TABLE 2 Overall comparison of single-stage YOLO variants, YOLO11n based configurations, and the proposed two-stage framework.

Method	Precision	Recall	F1-score	mAP@50
YOLO11n	0.83	0.88	0.85	0.89
YOLO11s	0.87	0.90	0.88	0.91
YOLOv8n	0.85	0.89	0.87	0.87
YOLOv8s	0.87	0.91	0.89	0.90
YOLOv8m	0.88	0.91	0.90	0.91
DCS-Attn	0.64	0.73	0.68	0.75
DCS-BHM	0.76	0.84	0.80	0.80
DCS-HN	0.87	0.82	0.84	0.89
Proposed	0.92	0.94	0.93	0.93

Among the configurations, it is observed that fine-tuned variants improve over the attention-base configuration. DCS-BHM increases the F1-score from 0.68 to 0.80 and improves mAP@50 from 0.75 to 0.80. This improvement indicates that exposing the model to difficult samples helps recover part of the performance lost in DCS-Attn. However, its precision remains relatively low at 0.76. This suggests that the balanced mining strategy still faces challenges with ambiguous samples that increase false positives. On the other hand, DCS-HN achieves higher precision of 0.87 and mAP@50 of 0.89. This shows that focusing on hard negatives can reduce

TABLE 3 Performance comparison of the proposed framework with representative two-stage detectors.

Method	Precision	Recall	F1-score	mAP@50
Faster R-CNN	0.75	0.71	0.73	0.70
Cascade R-CNN	0.88	0.89	0.88	0.88
Mask R-CNN	0.87	0.90	0.88	0.89
Libra R-CNN	0.89	0.91	0.90	0.90
Double-Head R-CNN	0.90	0.91	0.90	0.91
Proposed	0.92	0.94	0.93	0.93

false detections and improve confidence in positive predictions. However, its recall decreases to 0.82 which indicates that the model becomes more conservative and missed true defect instances.

By comparing the performance of single-stage detector with YOLO11n modified configurations, Table 2 shows that the performance of YOLO11n modified configurations achieve lower performance than standard single-stage YOLOn. This suggests that directly integrating the attention modules does not necessarily improve detection performance. These results indicate that integrating the attention mechanism has reduced the performance and increases the complexity with no obvious improvement in performance.

4.4 Comparison with two-stage models

Since the proposed framework is two-stage, therefore, its performance is compared with other two-stage frameworks. In this experiment, two-stage detectors are used such as Faster R-CNN (Ren et al., 2015), Cascade R-CNN (Cai and Vasconcelos, 2018), Mask R-CNN (He et al., 2017), Libra R-CNN (Pang et al., 2019), and Double-Head R-CNN (Wu et al., 2020). These detectors are used as competitive benchmarks in various object detection tasks due to their strong localization capability.

The results are reported in Table 3. From the Table 3, it is shown that Faster R-CNN achieves lowest performance compared to other methods with a precision of 0.75, recall of 0.71, F1-score of 0.73, and mAP@50 of 0.70. This is because that Faster R-CNN uses RPN for generating object proposal. However, Faster R-CNN cannot detect small objects because of the large receptive field. Cascade R-CNN performs better with F1 score of 0.88 and mAP@50 of 0.88. This improvement may be attributed to cascaded refinement strategy, which progressively improves localization quality. Similarly, Mask R-CNN achieves an F1-score of 0.88 and mAP@50 of 0.89, showing that the additional mask branch can support better spatial representation, although its advantage is limited. Libra R-CNN and Double-Head R-CNN achieve good performance than the earlier two-stage baselines. Libra R-CNN obtains a precision of 0.89, recall of 0.91, F1-score of 0.90, and mAP@50 of 0.90. This indicates that the balanced learning helps reduce the effect of sample and feature imbalance. Double-Head R-CNN achieves comparable performance, with a precision of 0.90, recall of 0.91, F1-score of 0.90, and mAP@50 of 0.91. Despite these improvements, the proposed framework achieves the best overall performance, with a precision of 0.92, recall of 0.94, F1-score of 0.93, and mAP@50 of 0.93. Compared with the strongest conventional two-stage baseline, Double-Head R-CNN, the proposed framework improves precision by 0.02, recall by 0.03, F1-score by 0.03, and mAP@50 by 0.02.

We also report the qualitative detection results of single- and two-stage detectors. These detectors include, Cascade R-CNN, YOLO11n, and proposed method in Figure 4. As depicted from Figure 4, Cascade R-CNN demonstrates comparatively weaker detection performance with more missed defect instances and false positives. In contrast, YOLO11n provides improved detection performance as it correctly detects a larger proportion of the target instances while producing fewer false positives and false negatives than Cascade R-CNN. On the other hand, the proposed method demonstrates good performance by accurately identifying more true defect instances while suppressing both missed detections and false alarms.

4.5 Category-wise performance analysis

To comprehensively evaluate the performance of the proposed model and compare its performance with different YOLO-based variants, category-wise evaluation was conducted. The results of this experiment are provided in the Table 4. Among the standard YOLO variants, YOLOv8m achieves good performance across all the categories. YOLOv8s and YOLO11s also provide competitive results. In contrast, the proposed framework achieves the best overall performance. This experiment provides further insights about category-wise performance of the models. From the Table 4, it is noticed that for the `No_issues` category, most of the models achieve strong performance. YOLOv8m achieves the highest precision of 0.98, while the proposed framework obtains a precision of 0.95 and a higher recall of 0.97. The F1-score of 0.96 is comparable to those of YOLO11s, YOLOv8s, and YOLOv8m. These results indicate the proposed method still classifies a small number of defective regions as the normal category.

For the `shell_Broken` category, YOLOv8m achieves the highest precision of 0.98 and an F1-score of 0.92. The proposed method obtains a lower precision of 0.94 but improves recall from 0.86 to 0.92, resulting in the highest F1-score of 0.93 for this category. For the `glaze_Flashover_damage` category, YOLOv8m achieves an F1-score of 0.96, while YOLOv8s obtains 0.95. The proposed framework achieves a precision of 0.93, recall of 0.94, F1-score of 0.94, and AP of 0.95. However, the results for `notbroken-notflashed` category show that it is the most challenging category for the standard YOLO detectors. YOLO11n, YOLO11s, YOLOv8n, YOLOv8s, and YOLOv8m achieving high recall values ranging from 0.91 to 0.97, but their precision remains between 0.53 and 0.61. These results show that these models detect most true instances of this category but also incorrectly classify many samples belonging to this category. Therefore, their F1-scores remain between 0.68 and 0.74. The proposed method due to two-stage pipeline improves the precision of 0.86 while maintaining a high recall of 0.93, resulting in an F1-score of 0.89 and AP of 0.88.

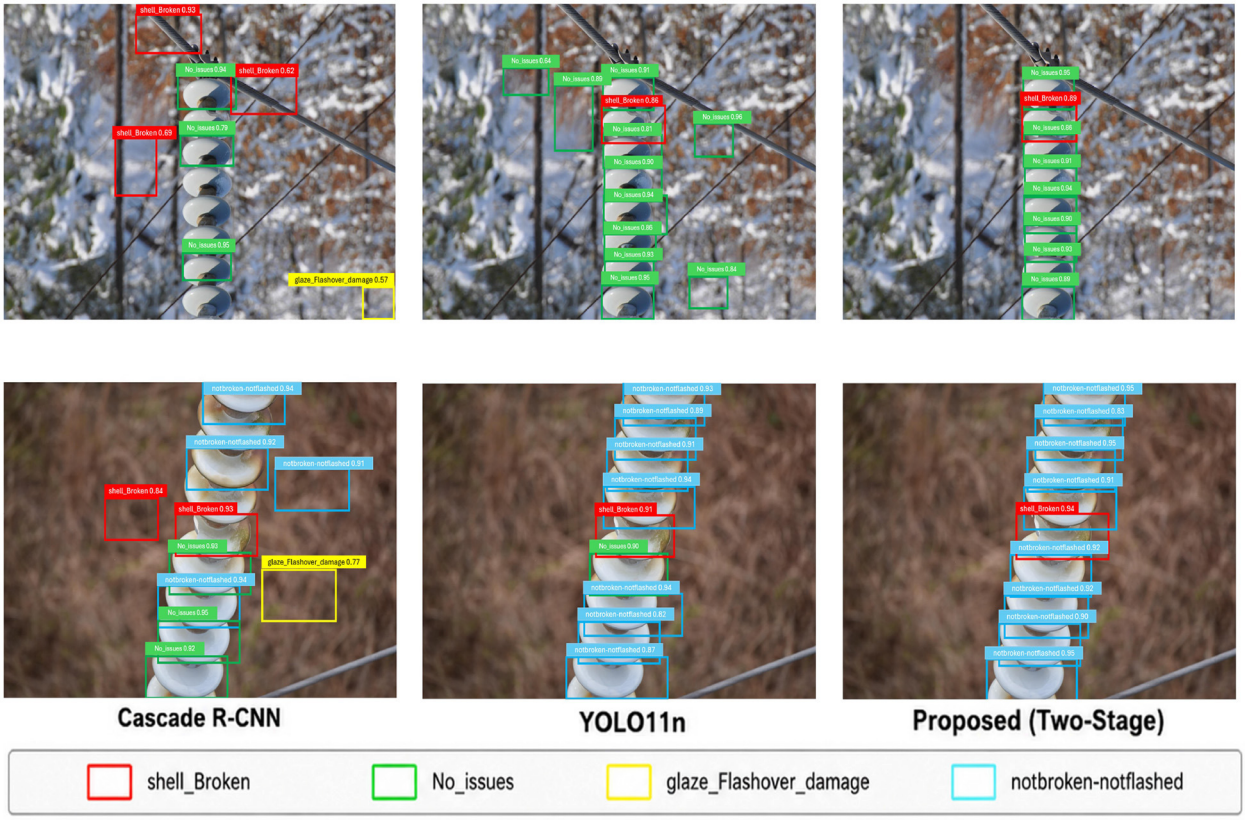


FIGURE 4
Illustrates qualitative detection result of different models. First column shows the prediction of Cascade R-CNN model, second column represents the results of YOLO11n, while the third column shows the results of the proposed framework.

4.6 Ablation study

Ablation study is performed to evaluate the effect of different components of proposed framework on overall performance.

4.7 Effect of first-stage detector

The ablation study evaluates how the various YOLO-based detectors (utilized in the first stage) affect the final performance of the two-stage framework. In this study, different variants of YOLO are used and the MobileNetV3 is kept as second-stage patch classifier. The results of this experiment are provided in Table 5. From the Table 5, it is indicated that the two-stage framework improves the performance of single-stage detector. It is also noticed that level of improvement depends on the performance of single-stage detectors which is responsible for generating the initial proposals/candidate regions. For example, the lightweight YOLOv8n and YOLO11n models achieve two-stage mAP@50 values of 0.89 and 0.90, respectively, while the stronger YOLOv8s, YOLO11s, and YOLOv8m detectors achieve better final results. YOLOv8m provides the strongest configuration, improving from a first-stage precision, recall, and mAP@50 of 0.88, 0.91, and 0.91 to 0.92, 0.94, and 0.93 after MobileNetV3 classifier. On the other hand, DCS-Attn performed relatively lower because its poor first-stage localization limits the quality of proposals. DCS-BHM partially improves the performance, whereas DCS-HN achieves a more balanced result by reducing false positives.

4.8 Second-stage classifier backbone ablation

To evaluate the effect of second-stage classification backbone on overall performance of two-stage framework, an ablation study with different backbone networks is conducted. Specifically, MobileNetV3, MobileNetV4-Conv-Small, ResNet-18, EfficientNet-B0, and ResNet-50 models are used in the second-stage while YOLO11n is kept fixed as the first-stage detector in all the experiments. The results of this ablation study are provided in Table 6.

As shown in Table 6, the performance increases with the increase in the size and capacity of the second-stage backbone. MobileNetV3 achieves the lowest performance, with a precision of 0.89, recall of 0.91, F1-score of 0.89, and mAP@50 of 0.90. This is due to the reason that MobileNetV3 is lightweight and computationally efficient network and has limited feature extraction capacity which reduces its ability to distinguish complex defect patterns. ResNet-18 improves the F1-score to 0.91 and mAP@50 to 0.92, while EfficientNet-B0 provides a stronger balance,

TABLE 4 Category-wise and overall performance comparison of single-stage YOLO variants, YOLO11n-based DCS configurations, and the proposed two-stage framework.

Model	Category	P	R	F1	AP
DCS-Attn	No_issues	0.83	0.95	0.88	0.94
	glaze_Flashover_damage	0.55	0.77	0.64	0.78
	notbroken-notflashed	0.38	0.55	0.45	0.49
	shell_Broken	0.81	0.65	0.72	0.80
DCS-BHM	No_issues	0.91	0.94	0.92	0.97
	glaze_Flashover_damage	0.83	0.85	0.84	0.92
	notbroken-notflashed	0.43	0.86	0.64	0.45
	shell_Broken	0.89	0.73	0.80	0.87
DCS-HN	No_issues	0.90	0.94	0.92	0.96
	glaze_Flashover_damage	0.84	0.78	0.81	0.89
	notbroken-notflashed	0.84	0.88	0.86	0.86
	shell_Broken	0.92	0.69	0.79	0.87
YOLO11n	No_issues	0.92	0.95	0.93	0.97
	glaze_Flashover_damage	0.94	0.81	0.87	0.94
	notbroken-notflashed	0.53	0.95	0.68	0.75
	shell_Broken	0.94	0.82	0.88	0.92
YOLO11s	No_issues	0.97	0.95	0.96	0.98
	glaze_Flashover_damage	0.97	0.91	0.94	0.98
	notbroken-notflashed	0.58	0.93	0.72	0.74
	shell_Broken	0.98	0.84	0.90	0.94
YOLOv8n	No_issues	0.96	0.92	0.94	0.98
	glaze_Flashover_damage	0.94	0.86	0.90	0.95
	notbroken-notflashed	0.57	0.97	0.71	0.65
	shell_Broken	0.97	0.81	0.89	0.92
YOLOv8s	No_issues	0.97	0.95	0.96	0.98
	glaze_Flashover_damage	0.97	0.93	0.95	0.98
	notbroken-notflashed	0.59	0.91	0.71	0.72
	shell_Broken	0.97	0.86	0.91	0.94
YOLOv8m	No_issues	0.98	0.94	0.96	0.98
	glaze_Flashover_damage	0.97	0.93	0.95	0.98
	notbroken-notflashed	0.61	0.94	0.74	0.74
	shell_Broken	0.98	0.86	0.92	0.97
Proposed	No_issues	0.95	0.97	0.96	0.97
	glaze_Flashover_damage	0.93	0.94	0.94	0.95
	notbroken-notflashed	0.86	0.93	0.89	0.88
	shell_Broken	0.94	0.92	0.93	0.92

P, Precision; R, Recall; F1, F1-score; and AP, mAP@50.

reaching a precision of 0.92, recall of 0.93, F1-score of 0.92, and mAP@50 of 0.92. ResNet-50 achieves the best overall performance, with a precision of 0.94, recall of 0.93, F1-score of 0.93, and mAP@50 of 0.94.

From this experiment, it can be concluded that the deeper backbone models provide more discriminative features for patch-level classification. However, these models also introduce high computation costs. Since the objective of this study is to develop lightweight solutions for defect detection suitable for UAV deployment, where onboard hardware resources are typically limited, the second-stage classifier should be selected based on balancing detection accuracy and computational efficiency.

TABLE 5 Ablation study of different YOLO variants used as the first-stage detector in the two-stage framework.

YOLO variant	Second stage	Single-stage detector			Two-stage framework		
		P	R	mAP@50	P	R	mAP@50
YOLOv8n	MobileNetV3	0.85	0.89	0.87	0.88	0.91	0.89
YOLO11n		0.83	0.88	0.89	0.89	0.91	0.90
YOLOv8s		0.87	0.91	0.90	0.90	0.92	0.91
YOLO11s		0.87	0.90	0.91	0.91	0.93	0.92
YOLOv8m		0.88	0.91	0.91	0.92	0.94	0.93
DCS-Attn		0.64	0.73	0.75	0.78	0.82	0.80
DCS-BHM		0.76	0.84	0.80	0.84	0.88	0.86
DCS-HN		0.87	0.82	0.89	0.90	0.90	0.90

MobileNetV3 is fixed as the second-stage classifier for all configurations.

TABLE 6 Ablation study on the second-stage classification backbone.

Model	P	R	F1	mAP@50	Params (M)	GFLOPs	FPS
MobileNetV3	0.89	0.91	0.89	0.90	2.54	0.06	30
MobileNetV4-Conv-Small	0.91	0.92	0.91	0.91	3.8	0.15	27
EfficientNet-B0	0.92	0.93	0.92	0.92	5.29	0.39	24
ResNet-18	0.91	0.92	0.91	0.92	11.69	1.8	22
ResNet-50	0.94	0.93	0.93	0.94	25.56	4.1	18

The first-stage detector (YOLO11n) is kept fixed for all configurations. Computational complexity is reported for the second-stage classifier.

TABLE 7 Accuracy–complexity comparison of different configurations.

Model	F1-score	mAP@50	Params (M)	FPS
YOLOv8n	0.87	0.87	3.01	11.2
YOLO11n	0.85	0.89	2.58	10.1
YOLO11s	0.89	0.91	9.41	9.6
YOLOv8m	0.90	0.91	25.84	9.8
Proposed two-stage framework	0.930	0.930	28.40	8.3

For the proposed framework, the parameter count includes both YOLOv8m and MobileNetV3, while FPS represents end-to-end two-stage inference.

4.9 Computational complexity

In this study, the computational complexity of the proposed two-stage framework is compared with other single-stage models. This is important to evaluate the suitability of the proposed framework for UAV-based inspection and to determine whether the detection improvement justifies computational costs. The results are reported in Table 7. From the Table 7, YOLOv8n and YOLO11n achieve lowest computational complexity, with 3.01 M and 2.58 M parameters, respectively, and achieve the highest inference speeds of 11.2 and 10.1 FPS. However, F1-score of these models remain below 0.88.

YOLO11s improves the F1-score to 0.890 and mAP@50 to 0.910, but the inference speed decreases to 9.6 FPS. YOLOv8m achieves good performance compared to single-stage detectors, however, its computational complexity increases to 25.84 and FPS decreases to 9.8. The proposed framework achieves highest performance in terms of F1-score and mAP@50 with moderate increase in model size from 25.84 M to 28.40 M parameters and reduction in inference speed to 8.3 FPS.

4.10 Grad-CAM-based interpretation

To visually examine whether the proposed two-stage detection framework focuses on the meaningful insulator regions during prediction, the Grad-CAM experiment is performed. Grad-CAM is a tool used to evaluate the spatial attention behavior of the model. If the Grad-CAM maps are more concentrated around the insular defect regions indicate that the model is learning relevant object-level and defect-sensitive features rather than focusing on background information.

Figure 5 illustrates the Grad-CAM maps of DCS-Attn, YOLO11n, and proposed framework. The first column of the Figure 5 presents the sample input images, while the subsequent columns show the corresponding activation maps. For DCS-Attn (second column), the activations are dispersed across the image with several responses appearing on background regions rather than concentrated on the insulator. This shows that the model is influenced by irrelevant contextual features and cannot capture the compact representation of the target region. The Grad-CAM responses of YOLO11n (third column) show noticeable improvement with considerable responses around the insulator regions, however, some activations are distributed over the background and other regions. The response of proposed framework (fourth column) is more localized with activations concentrated around the insulator regions.

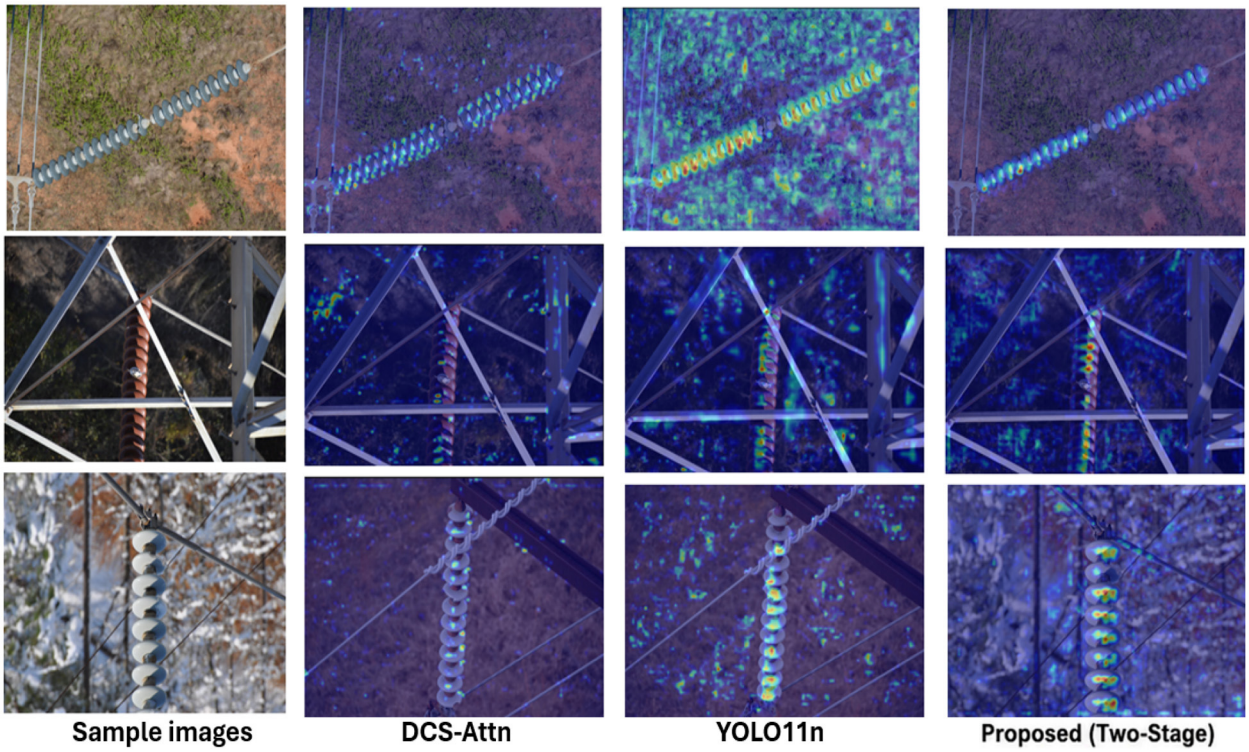


FIGURE 5
Illustrates Grad-CAM results of different methods. The first column shows the original input samples, the second column shows the Grad-CAM activation maps of DCS-Attn method, the third column show the activation maps of YOLO11n method, and the fourth column shows the Grad-CAM responses of proposed framework.

5 Conclusion

This study presents coarse-to-fine two-stage deep learning framework for insulator defect detection in UAV imagery. The proposed method separates the tasks of localization and classification into two complementary stages. The first stage utilizes YOLO-based detector to identify the candidate regions with coarse prediction. The second stage applies lightweight patch-level classifier that classifies the patches corresponding to candidate regions. Furthermore, to improve inter-class separability, HPCCL is introduced in the embedding space. The proposed framework is comprehensively evaluated on public IDID. Experiment results demonstrate the effectiveness of proposed framework by achieving best overall performance compared with standard YOLO variants and two-stage frameworks.

Although the proposed framework is effective in detecting insulator defects in complex scenes, however, there are several limitations needed to be addressed. The performance of the overall framework depends upon the quality of YOLO-based detector in the first stage. Detection missed in the first stage cannot be recovered in the second stage. Furthermore, the current evaluation is performed on single dataset, and further validation is required on large multi-source datasets. The two-stage pipeline also introduces additional computational overhead due to patch extraction and classification. This limits the deployment of the proposed model on UAVs with constrained resources. In future, these limitations will be addressed by validating the model on diverse dataset and optimize the model for real-time deployment.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author.

Author contributions

SK: Investigation, Conceptualization, Software, Writing – review & editing, Data curation, Methodology, Writing – original draft, Formal analysis, Visualization. AN: Methodology, Formal analysis, Writing – review & editing, Project administration. HU: Formal analysis, Project administration, Funding acquisition, Writing – review & editing, Data curation, Visualization. MU: Resources, Funding acquisition, Writing – review & editing, Formal analysis, Data curation.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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